Answer ALL questions.

MODULE 1: FUNDAMENTALS OF CHEMISTRY

1. (a) Figure 1 shows the electron transitions which occur between the energy levels in the hydrogen spectrum.

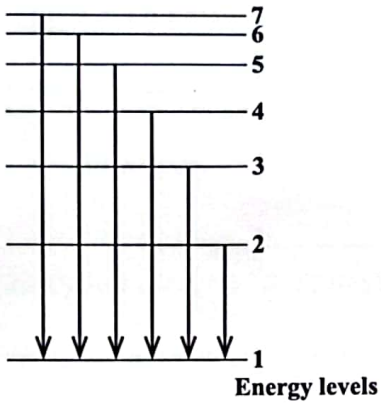

Figure 1. Electron transitions between the energy levels in the hydrogen spectrum

Name the series of lines produced by these electron transitions.

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[1 mark]

(b) Draw diagrams (including the axes) to show the shapes of the **s** and **p** orbitals.

[2 marks]

(c) In methane, the carbon atom uses four electrons in hybridized orbitals to bond to four hydrogen atoms. What is meant by the term 'hybridization'?

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[2 marks]

(d) Benzene exhibits resonance structures. What is meant by 'resonance'?

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[2 marks]

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(e) (i) State THREE criteria used in the valence shell electron pair repulsion (VSEPR) theory to determine the shapes of molecules and ions.

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[3 marks]

(ii) Use the criteria in (e) (i) to determine the shape **and** bond angles in the

a) beryllium chloride molecule

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[2 marks]

b) methyl anion, CH_3^-.

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[2 marks]

(iii) Draw a diagram to show the shape of the methyl anion, CH_3^-.

[1 mark]

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(f) Potassium nitrate, KNO_3, decomposes on heating as shown in the equation below.

$$2\ KNO_3(s) \longrightarrow 2\ KNO_2(s) + O_2(g)$$

A sample of KNO_3 weighing 6.20 g (Sample A) was partly decomposed on heating. The residue was dissolved in water and made up to 1 dm^3 of solution. Then, 25.0 cm^3 of the solution was pipetted into a conical flask and acidified with an equal volume of dilute sulfuric acid. The mixture was titrated with 0.02 mol dm^{-3} potassium manganate(VII) solution and an average titre was found to be 27.50 cm^3.

(i) Calculate the number of moles of potassium manganate(VII) used in the titration.

[1 mark]

(ii) Using the equation,

$$2\ MnO_4^-(aq) + 5\ NO_2^-(aq) + 6\ H^+(aq) \longrightarrow Mn^{2+}(aq) + 5\ NO_3^-(aq) + 3\ H_2O(l)$$

Deduce the number of moles of KNO_2 in the titre volume.

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[1 mark]

(iii) Calculate the number of moles of KNO_2 that was present in the residue from the decomposition of Sample A.

[1 mark]

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(iv) Calculate the mass of KNO_2 present in the residue from Sample A.

[2 marks]

(v) Using the equation for the decomposition of KNO_3, determine the mass of KNO_2 formed if 6.20 g KNO_3 were completely decomposed.

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[3 marks]

(vi) Using the results from (f) (iv) and (v), calculate the percentage of KNO_2 formed.

[2 marks]

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(vii) Outline the steps taken in the laboratory to prepare the 0.02 mol dm^{-3} potassium manganate(VII) solution used for this titration.

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[5 marks]

Total 30 marks

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MODULE 2: KINETICS AND EQUILIBRIA

2. (a) What is meant by the term 'dynamic equilibrium'?

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[2 marks]

(b) The reaction between A and B to form C and D is represented in the following equation.

$$A(l) + B(l) \rightleftharpoons C(l) + D(l)$$

The equilibrium constant value for the reaction is 4.0.

(i) Write an expression for the equilibrium constant of the **reverse** reaction.

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[1 mark]

(ii) State the numerical value of the equilibrium constant for the reverse reaction based on the expression in (b) (i).

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[1 mark]

(c) State Le Chatelier's principle.

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[2 marks]

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(d) Consider the following equilibrium reactions.

Equilibrium A: $H_2O(g) + C(s) \rightleftharpoons H_2(g) + CO(g)$ $\Delta H = +131\ kJ\ mol^{-1}$

Equilibrium B: $2CrO_4^-(aq) + 2H^+(aq) \rightleftharpoons Cr_2O_7^{2-}(aq) + H_2O(l)$

Use Le Chatelier's principle to predict and explain the effect of

(i) increasing the pressure on Equilibrium A

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[2 marks]

(ii) decreasing the temperature on Equilibrium A

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[2 marks]

(iii) increasing $[H^+(aq)]$ on Equilibrium B.

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[2 marks]

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(e) A buffer solution containing 0.2 mol dm^{-3} of a weak acid, HA, and 0.5 mol dm^{-3} of the sodium salt of the weak acid, NaA, was prepared for use in the food industry.

(i) Define the term 'buffer solution'.

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[2 marks]

(ii) Given that Ka for the weak acid = 7.4×10^{-4} mol dm^{-3}, calculate the pH of the buffer solution.

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[3 marks]

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(f) Outline the experimental procedure which can be used to prepare a sample of the buffer solution in (e). Include in your answer tests to be used to ensure a buffer solution was made.

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[5 marks]

(g) With the aid of equations, explain how a solution of ammonium sulfate and ammonia can control pH.

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[4 marks]

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(h) Define the term 'half-life'.

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[2 marks]

(i) Using a Boltzmann distribution curve, show how a catalyst affects the rate of a chemical reaction.

[2 marks]

Total 30 marks

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MODULE 3: CHEMISTRY OF THE ELEMENTS

3. (a) (i) Write the formulae of the oxides of the elements of the third period, from sodium to sulfur.

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[3 marks]

(ii) Explain the difference in the oxidation number in the oxides of sodium and aluminium.

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[3 marks]

(iii) The type of bond in the chlorides of Period 3 elements depends on the electronegativities of the atoms in the bond. State the type of bond present in the following molecules.

a) Magnesium chloride

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b) Silicon chloride

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c) Phosphorus(III) chloride

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[3 marks]

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(iv) With the aid of an equation, describe the reaction of silicon chloride in water.

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[4 marks]

(b) A transition element ion, X^{3+}, forms a complex with six hydroxide ligands.

(i) Define the term 'ligand'.

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[2 marks]

(ii) Write the formula of the complex formed.

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[1 mark]

(iii) State the shape of the complex formed.

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[1 mark]

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(iv) Using a suitable diagram, illustrate your answer in (b) (iii).

[2 marks]

(v) State what is meant by the term 'coordination number', in reference to a complex ion.

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[2 marks]

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(c) (i) State the expected observation for the addition of concentrated hydrochloric acid to aqueous copper(II) sulfate.

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[2 marks]

(ii) Write an equation for the reaction which occurs and identify the species with the higher K_{stab} (stability constant) value.

[3 marks]

(d) Acidified potassium manganate(VII) solution is a strong oxidizing agent. It can be used in a redox titration to determine the concentration of Fe^{2+} in iron tablets.

(i) Describe the changes which occur at the end point of the titration?

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[2 marks]

(ii) Write the formula and state the colour of the species formed from the oxidation of Fe^{2+} ions.

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[2 marks]

Total 30 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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